MILITARY SPECIFICATION

PROJECTILE, GUIDED,
155MM, XM712, WARHEAD FOR

This amendment forms a part of Military Specification MIL-P-63226(AR), dated 1 August 1979, and is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "M712" to title and wherever it appears in this specification.

Specifications, Military: Add, "MIL-C-450C Coating Compound Bituminous Solvent Type".

PAGE 2

DRAWINGS: Delete the following:

"9300922 - "Adhesive, Explosive Compatible Epoxy, Two Component
"9307009 - Seal, End Closure
"9307046 - Wiring Harness,Warhead - 3W1 - 3W2".

DRAWINGS: Add the following:
"9300374 - Filler, Molded
"9300786 - Adhesive, Explosive Compatible
"9300790 - Adhesive, Epoxy, Two Component
"9300792 - Powder Fluorescent
"9300793 - Release Agent
"9307062 - Seal, Cable Exit
"9307069 - Wiring Harness"

Delete the following in its entirety:

"Equipment Operating Instructions

"EOI 9304456 - Warhead Test Set, Equipment Operating Instruction"

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1320

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PAGE 3

3.2.1.1, line 8: Delete "9307046 and substitute "9307069"

3.2.1.2, line 8: Delete "9307046" and substitute "9307069". Delete the following: "This requirement does not include fuze wiring.".

3.2.1.3, line 2: Delete "installation" and substitute "electrical connection".

line 3: Add "The electrical checks per Table 1 are not required, at the warhead section level, if performed as part of the warhead section/control section cable resistance checks."

PAGE 4

3.2.1.4, line 2: Delete "10^{-4}" and substitute "10^{-2}".

PAGE 6

3.2.1.5.7.2(b): Delete in its entirety and substitute the following:

"3.2.1.5.7.2(b) Cavitation, cracks or annular rings in segments A or B in excess of that permitted by the applicable requirement, but not in excess of twice that permitted, except that such cavities when located more than 1.00 inch from the warhead axis in segment A shall be disregarded."

PAGE 7

Add the following paragraphs:

"3.2.1.9 Bonding aft harness.

"3.2.1.9.1 After bonding the harness to 4.03 diameter surface, assure smooth, void and air bubble free adhesive in slots and under and around cables.

"3.2.1.9.2 After bonding glass cloth tape strips to 4.03 diameter, assure smooth continuous outer surface of adhesive and smooth continuous fillets covering all edges of the glass cloth tape. No voids, bubbles, or adhesive starved areas shall be visible.

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"3.2.1.9.3 After bonding the metal shims assure no voids, discontinuities of air bubbles are visible along any edge of shim."

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* 4.4.2.2: Delete in its entirety and substitute the following:

"4.4.2.2 Lot Acceptance - Lot acceptance shall be based on acceptance of individual sublots consisting of warheads poured from explosive melted at one time in a single melt kettle. Beginning with the first sublot produced and continuing until five consecutive sublots have met the requirements of drawing 9307058 (Confidential), two units shall be selected at random from each sublot for static penetration in accordance with 4.5.8. After five consecutive sublots have met the requirements of drawing 9307058 (Confidential), one unit shall be selected at random from each sublot for static penetration in accordance with 4.5.8. Failure of any loaded warhaed assembly to meet the requirements of drawing 9307058 (Confidential) shall be cause for rejection of the sublot and processing in accordance with 4.4.3.2. Sublot acceptance tests shall be performed in accordance with Table VI."

* 4.4.3.2: Delete in its entirety ans substitute the following:

"4.4.3.2 Sublot rejection - Any sublot failing to meet the requirements of 4.4.2.2 shall be resubmitted once by selecting two additional loaded warhead assemblies for tests. This test set-up shall make provisions for flash x-ray to record radiographs of the shaped charge jet in two planes at 90 degrees apart. If both loaded warhead assemblies meet the requirements of drawing 9307058 (Confidential), the sublot shall be accepted. Failure to meet the requirements of drawing 9307058 (Confidential) shall cause the sublot to be rejected and the provisions of 4.4.2.2. shall again apply until five consecutive sublots are accepted. Rejected sublots, with test data and radiographs, shall be submitted to the procuring activity for disposition."

PAGE 10

4.5.3: Delete in its entirety and substitute the following:

"4.5.3 Leak test. The empty warhead assembly shall be pressurized to 15 $\pm$ 1.0 psig. The leakage rate shall not exceed the equivalent of 10^{-2} cc/sec of air at standard conditions. A leak rate of 10^{-4} cc/sec of air at standard conditions may be detected by immersion in water or other clear liquid for 30 seconds minimum. A continuous stream of bubbles will be emmitted if the leak exceeds 10^{-2} cc/sec."

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4.5.4.3: Delete in its entirety and substitute the following:

"4.5.4.3 Warhead section test. The electrical characteristics of the warhead section shall be tested and shall meet the requirements of Table I. If the fuze module was mechanically installed in the warhead prior to the time of the tests of 3.2.1.2 were conducted, it is not necessary to retest conductors 6 thru 18 in Table I. Resistance measurements shall be made with the positive output of the meter connected to the "From" conductor and the negative output of the meter connected to the "To" conductor of Table I. The test current shall be limited to 1 milliamp or less. The electrical checks per Table 1 are not required, at the warhead section level, if performed as part of the warhead section/control section cable resistance checks. However, if the warhead is supplied as GFM, then all of the tests in Table I are applicable."

PAGE 16

Add the following paragraph:

"4.5.12 Explosive leakage inspection. The warhead shall be inspected to verify conformance to the non-leakage requirement of 3.2.1.6."

PAGE 22

Critical 1, 2, and 3: Delete in their entirety and substitute the following:

"1	No explosive material, other than within charge cavity	100%	3.2.1.6	4.5.12
"2	Warhead cavity finish/ cleanliness	100%	3.2.1.7	4.5.9
"3	Defects in explosive charge	100%	3.2.1.5.7.1	4.5.5 4.5.6"

PAGE 25

Major 108: Delete in it's entirety and substitute the following:

"108	Distance between face of liner and datums A and B	0.40%	Dwg	Gage"

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Critical: Delete the following in their entirety:

"1	Thickness	100%	Dwg.	Gage
"2	Material	NA	Dwg	Vendor Lot Certification
"3	Hardness	100%	Dwg	Gage"

Major: Add the following:

"101	Thickness	100%	Dwg	Gage
"102	Material	NA	Dwg	Vendor Lot Certification
"103	Hardness	100%	Dwg	Gage"

PAGE 33

Critical: Delete the following in their entirety:

"1	Inspection of weld and heat treatment, including cracks.	100%	Dwg	9300967
"2	Material	100%	Dwg	9300967
"3	Dry lube on groove surface	100%	Dwg	Visual
"4	Runout of aft interface shoulder to forward and aft datums	100%	Dwg	Gage
"5	Length of aft interface cylinder	100%	Dwg	Gage
"6	Location of lock ring groove from aft interface shoulder	100%	Dwg	Gage"

Critical: Add the following:

"1	Inspection of weld	100%	Dwg	9300967
"2	Heat treatment, yield and ultimate strength	Dwg	Dwg	9300967
"3	Inspection for cracks	100%	Dwg	9300967"

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Major: Add the following:

"111	Material	100%	Dwg	9300967
"112	Dry lube on groove surface	100%	Dwg	Visual
"113	Runout of aft interface shoulder to forward and aft datums	100%	Dwg	Gage
"114	Length of aft interface cylinder	100%	Dwg	Gage
"115	Location of lock ring groove from aft interface shoulder	100%	Dwg	Gage
"116	Heat treatment, elongation	Dwg	Dwg	9300967"

PAGE 38

Critical: Delete the following in their entirety:

"1	No unsolidified paint in charge cavity	100%	3.2.1.7	4.5.9
"2	Finished surfaces	100%	3.2.1.7	4.5.9
"3	Adhesive	NA	Spec.	9300922 - Vendor lot Certification"

Critical: Add the following:

"1	Adhesive explosive compatibility	NA	Dwg	9300786
"2	Bond of aft harness	100%	3.2.1.9.1	Visual
"3	Bond of glass cloth	100%	3.2.1.9.2	Visual
"4	Bond of metal shim	100%	3.2.1.9.3	Visual
"5	Adhesive	NA	Spec.	9300790 - Vendor lot Certification"

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PAGE 39

Critical 1, 2, and 3: Delete in their entirety and substitute the following:

"1	No explosive material, other than within charge cavity	100%	3.2.1.6	4.5.12
"2	Warhead cavity finish/ cleanliness	100%	3.2.1.7	4.5.9
"3	Defects in explosive charge	100%	3.2.1.5.7.1	4.5.5 4.5.6"

PAGE 41

Critical 1,2, and 3: Delete in their entirety and substitute the following:

"1	No explosive material other than within charge cavity	100%	3.2.1.6	4.5.12
"2	Warhead cavity finish/ cleanliness	100%	3.2.1.7	4.5.9
"3	Defects in explosive charge	100%	3.2.1.5.7.1	4.5.5 4.5.6 Review existing radio-graphs"

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